

Homework Assignment - 3

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part - c

Problem C.1

Data Dependents Table

Register	Source Instruction	Dependent Instruction	Type
X1	ld X1, 0(X2)	addi X1, X1, 1	RAW
X1	addi X1, X1, 1	sd X1, 0(X2)	RAW
X2	ld X1, 0(X2)	addi X2, X2, 4	Loop Carried RAW
X2	sd X1, 0(X2)	addi X2, X2, 4	L.C. RAW
X2	Sub X4, X3, X2	addi X2, X2, 4	L.C. RAW
X4	Sub X4, X3, X2	bnez X4, Loop	RAW

② X1 has two RAW dependences within the loop body

ld $\rightarrow$ addi & addi $\rightarrow$ sd

③ X2 is updated in each loop and carries dependences across iterations.

④ X4 is produced by Sub and used by bnez for branch control.

⑤ X3 is constant, so there is no dependence hazard.

Problem C.2

Considering a classic 5-stage pipeline

IF $\rightarrow$ ID $\rightarrow$ EX $\rightarrow$ MEM $\rightarrow$ WB

Identifying all true RAW hazards

Register	Dependent Ins.	Depends on	Type.
x1	addi x1, x1, 1	ld x1, 0(x2)	load-use hazard
x1	sd x1, 0(x2)	addi x1, x1, 1	ALU-Store hazard
x2	sub x4, x3, x2	addi x2, x2, 4	ALU-ALU hazard
x4	bne2 x4, loop	sub x4, x3, x2	branch hazard.

load-use hazard (2 stall cycles)

addi $\rightarrow$ sd (1 stall)

addi x2 $\rightarrow$ sub (1 stall)

Sub $\rightarrow$ bnez (1 stall)

Branch flush (2 instruction slots wasted).

Pipeline Timing Chart:-

[illegible]

overlapped.

$$\begin{aligned}\text{Total cycles per instruction} &= 6 \text{ instruction} \times 5 \text{ stages (10 cycles)} \\ &+ \text{stall (5 cycles)} + \text{Branch flush (2 cycles)} \\ &= 17 \text{ cycles. //}\end{aligned}$$

C.7

a) To calculate the speedup of the 12-stage pipeline over the 5-stage pipeline, considering only data hazards.

5 stage pipeline $\rightarrow$ clock cycle time (1 ns), Data hazard stall (1 stall every 5 instruction)

12 stage pipeline $\rightarrow$ clock (0.6 ns), Data Hazard (3 stall every 8 instruction)

calculating CPI

$$\text{stall freq} = 1/5 = 0.2$$

every stall adds 1 extra cycle.

$$\text{CPI}_5 = 1 + 0.2 = 1.2$$

11th for 12 stage pipeline

$$\text{stall freq} = \frac{3}{8} = 0.375$$

$$\text{CPI}_{12} = 1 + 0.375 = 1.375$$

Avg. execution time per instruction (CPI / clock cycle time)

$$5 \text{ stage} \rightarrow \frac{1.2}{1} = 1.2$$

$$12 \text{ stage} \rightarrow \frac{1.375}{0.6} = 0.825$$

$$\text{Speedup} = \frac{\text{performance of 12-stage}}{\text{performance of 5 stage}}$$

$$= \frac{1.2}{0.825} = 1.4545 //$$

$\therefore$ Speedup is $1.45\times$ (i.e 45% faster)

b). Branch misprediction penalty for 5 stage (2 cycles) and 12 stage (5 cycles)

for 5 stage pipeline, branch freq = 0.20 and misprediction rate = 0.05

$$f_{\text{mispredict}} = (\text{branch freq}) \times (\text{misprediction rate})$$

$$= 0.20 \times 0.05 = 0.01$$

$$CPI_{\text{total}} = \text{base CPI} + f_{\text{mispredict}} \times \text{penalty}$$

$$CPI_5 = 1.2 + 0.01 \times 2 \Rightarrow 1.22$$

$$CPI_{12} = 1.375 + 0.01 \times 5 \Rightarrow 1.425$$

from this we can conclude that the deeper pipelines suffer more CPI degradation.

Question 3.1

operation	Latency beyond 1	Total cycles of Complete.
fld	+3	4
fmul.d	+4	5
fdiv.d	+10	11
fld	+3	4
fadd.d	+2	3
fadd.d	+2	3
fsub	+1	2
addi	-	1
addi	-	1
sub	-	1
bnz	+1	2

Sum total cycles per iteration = 37 cycles.

Question 3.2

fld f2, 0(Rx)	1+3
fmul.d f2, f0, f2	1+4
fdiv.d f2, f2, f0	1+10
fld f4, 0(Ry)	1+3
fadd.d f4, f0, f4	1
fadd.d f10, f2, f2	
fsub f4, 0(Ry)	

addi Rx, Rx, 8

addi Ry, Ry, 8

sub x20, x4, Rx

bnz x20, loop

Total cycles for the loop body is $= 4 + 5 + 11 + 4 + 1$
 $= 25$ cycles //

Question 3.3

For this we now have a dual issue pipeline, full forwarding and the stalls are to respect true RAW dependencies. The number of cycles required by the loop is 22. The execution starts with the fld instruction, followed by fdiv.d, that is supposed to wait four extra cycles to finish fld. Then fmul.d runs in ||el as fdiv.d because there is no dependency between them, although both need same input and have to wait to be ready.

Question 3.4

Resulting Code: (Renamed using T registers T9, T10.)

Loop: fld T9, 0(Rx)

I0: fmul.d T10, F0, T9

I1: fdiv.d T11, F0, T9

I2: fld T12, 0(Ry)

I3: fadd.d T13, F0, T12

I4: fadd.d T14, T11, T9

I5: sd T12, 0(R4)

Thus we have renamed the code using T registers, while preserving all true data dependences.

Question: 3.18

Given:

$$\text{Branch freq} = 15\% = 0.15$$

$$\text{BTB (hit rate)} = 0.9$$

$$\text{accuracy} = 0.9$$

$$\text{Speedup} = \frac{\text{CPI (no BTB)}}{\text{CPI (BTB)}} = \frac{\text{CPI (base)} + \text{stall (base)}}{\text{CPI (base)} + \text{stall (BTB)}}$$

$$\text{stall (no BTB)} = 15\% * 2 \Rightarrow 30\% = 0.30$$

Calculating stalls.

1) BTB Miss

$$\text{Freq} = 15\% \times 10\% = 1.5\% \text{ per instruction}$$

$$\text{penalty} = 3 \text{ cycles}$$

2) BTB Hit (Correct prediction)

$$\text{Freq} = 15\% \times 90\% \times 90\% = 12.1\% \text{ per instruction}$$

$$\text{penalty} = 0 \text{ cycles}$$

3) BTB (Incorrect prediction)

$$\text{Freq} = (15 \times 90 \times 10)\% = 1.3\%$$

$$\text{penalty} = 4 \text{ cycles}$$

$$\text{Speedup} = \frac{\text{CPI (base)} + \text{stall (base)}}{\text{CPI (base)} + \text{stall (BTB)}}$$

$$= \frac{1.0 + 0.30}{1.0 + 0.097}$$

$$\text{Speedup} = 1.2$$